

The Primordial Quark Nucleus ($\mathcal{Q}_{raw}$): A Phase Transition and Thermodynamic Approach to Cosmological Genesis

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Expanded via Quantum Criticality Framework

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Abstract

This paper expands upon the Primordial Quark Theory, which posits that the pre-Big Bang universe existed as a unified, zero-volume raw matter nucleus ($\mathcal{Q}_{raw}$). By applying thermodynamic principles, quantum degeneracy pressure, and charge saturation, we re-interpret the emergence of fundamental particles. The dissociation of this nucleus at a critical energy state (E_{BB}) is modeled as a phase transition, where leptons (electrons) are emitted as physical derivatives of the raw quark mass. Furthermore, we propose that the subsequent spatial expansion and the formation of the Quark-Gluon Plasma (QGP) represent the concurrent birth of spacetime and the primordial "Cosmic Smoke".

1 Introduction

The standard cosmological model assumes a singularity of infinite density. The Primordial Quark Theory introduces a tangible state to this singularity: the unified "raw matter" ($\mathcal{Q}_{raw}$). In this state, spatial dimensions were non-existent, and all forces were unified. The Big Bang is thus characterized not as an explosion in existing space, but as a critical quantum rupture of this raw matter nucleus.

2 The Big Bang Dissociation Equation

Upon reaching critical charge saturation and quantum degeneracy limits, the primordial nucleus underwent a massive dissociation. The energy of the Big Bang (E_{BB}) acted as the catalyst for this critical phase transition:

$$\mathcal{Q}_{raw} \xrightarrow{E_{BB}} \sum_{i=1}^n q_i + \sum_{j=1}^m e_j^- \quad (1)$$

Where q_i represents the resulting fractional-charged quarks, and e_j^- represents the derived electrons with an independent negative charge.

3 Thermodynamic Derivation of the Electron

In contrast to the Standard Model, this framework classifies the electron not as an initially independent particle, but as a physical derivative of the primordial matter during cosmic expansion. Mathematically, this transition is expressed as a partial derivative with respect to the expansion energy:

$$e^- = \frac{\partial(\mathcal{Q}_{raw})}{\partial(E_{BB})} \quad (2)$$

Thermodynamically, this represents the particle yield at the critical threshold. The rupture of $\mathcal{Q}_{raw}$ necessitated the emission of negatively charged electrons to conserve the overall charge balance against the positively fractional quarks, resolving the unified symmetry.

4 Thermodynamics of the "Cosmic Smoke" (Quark-Gluon Plasma)

Immediately following the dissociation of $\mathcal{Q}_{raw}$ by E_{BB} , the universe entered a state of an ultra-relativistic, opaque fluid known as the Quark-Gluon Plasma (QGP). This highly dense, hot state is conceptually analogous to primordial "Cosmic Smoke".

The thermodynamic energy density ρ_{QGP} of this early universe can be expressed using the Stefan-Boltzmann equation for a gas of relativistic particles:

$$\rho_{QGP} = \frac{\pi^2}{30} g_* \frac{(k_B T)^4}{(\hbar c)^3} \quad (3)$$

where T is the temperature and g_* represents the total effective degrees of freedom for the dissociated elementary particles (quarks, gluons, and the newly derived electrons). The degrees of freedom are defined by bosons and fermions:

$$g_* = \sum g_{bosons} + \frac{7}{8} \sum g_{fermions} \quad (4)$$

The outward thermodynamic pressure P driving the initial expansion (the "rupture") of this smoke is directly proportional to its energy density:

$$P = \frac{1}{3} \rho_{QGP} \quad (5)$$

As the primordial smoke expanded, spatial dimensions emerged dynamically. The expansion and cooling rate of this smoke is governed by the Friedmann equation (where H is the Hubble parameter and a is the scale factor):

$$H = \frac{\dot{a}}{a} = \sqrt{\frac{8\pi G}{3c^2} \rho_{QGP}} \quad (6)$$

When the temperature dropped below the critical hadronization threshold ($T < T_c \approx 150$ MeV), the "smoke" began to clear. The fractional-charged quarks (q_i) became confined into hadrons, while the derived electrons (e^-) persisted as stable leptons.

5 Conclusion

By treating the initial singularity as a primordial quark nucleus ($\mathcal{Q}_{raw}$), the Big Bang can be understood as an ultimate thermodynamic phase transition. The formulation $e^- = \frac{\partial(\mathcal{Q}_{raw})}{\partial(E_{BB})}$ provides a novel bridge between quantum criticality, the thermodynamic formation of the cosmic smoke (QGP), and the macroscopic genesis of spacetime.